

STAT GU4205/GR5205 Homework 1 [100 pts]
Due: Monday, October 5th at 11:59pm (ET)

Problem 1

Consider rolling two fair six sided dice (D_1 & D_2) and recording the sum of the faces and the maximum of the faces. Define two random variables $Y = D_1 + D_2$ and $X = \max\{D_1, D_2\}$. The joint probability distribution $P(X = x, Y = y)$ of these two random variables is provided in the lecture notes.

Perform the following tasks:

- 1.i Compute the true conditional expectations:

$$E[Y|X = x], \quad x = 1, 2, 3, 4, 5, 6$$

- 1.ii Use **R** or **Python** to simulate $n = 1000$ draws of (X, Y) . Plot the $n = 1000$ draws.

- 1.iii Assuming the conditional expectation takes on a linear form

$$E[Y|X = x] = \beta_0 + \beta_1 x,$$

estimate β_0 and β_1 using ordinary least squares. Include this line on your scatter plot from Part 1.i.

- 1.iv Predict the true conditional expectations $E[Y|X = x]$ using the estimated line $\hat{\beta}_0 + \hat{\beta}_1 x$. Do you believe that these estimates are biased?

- 1.v Estimate the bias of $\hat{Y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ for estimating the true conditional expectation $E[Y|X = x]$. For example, the true bias at $x = 3$ is

$$B(3) = E[\hat{Y}(3)] - E[Y|X = 3].$$

Your goal is to estimate $B(x)$, for $x = 1, 2, 3, 4, 5, 6$ using a simulation.

Problem 2

Consider bivariate normal random vector (X, Y) having joint probability density function:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right] \right\},$$

with $-\infty < x < \infty$, $-\infty < y < \infty$. Derive the conditional density of $Y|X = x$, i.e., simplify the expression

$$f(y|X = x) = \frac{f(x, y)}{f_X(x)},$$

where $f(x, y)$ is the joint density of X and Y and $f_X(x)$ is the marginal density of X . Note that $f_X(x) > 0$ for all x .

Problem 3 (1.22 KNN)

Sixteen batches of the plastic were made, and from each batch one test item was molded. Each test item was randomly assigned to one of the four predetermined time levels, and the hardness was measured after the assigned elapsed time. The results are shown below; X is the elapsed time in hours, and Y is hardness in Brinell units. Assume the first-order regression model (1.1) is appropriate (**model (2.1) in the notes**).

Data not displayed

Perform the following tasks:

- 3.i Use R to obtain the estimated regression function.
- 3.ii Use R to create a scatter plot with the line of best fit. Make the line of best fit red.
- 3.iii Use R to calculate the point estimate of σ^2 , i.e., compute MSE .
- 3.iv Use R to calculate the sample correlation coefficient and coefficient of determination.

Problem 4

Recall the sample residual is defined by $e_i = y_i - \hat{y}_i$, where y_i is the i th response value and $\hat{y}_i$ is its corresponding fitted value computed by least squares estimates $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Prove the following properties:

4.i

$$\sum_{i=1}^n x_i e_i = 0$$

4.ii

$$\sum_{i=1}^n \hat{y}_i e_i = 0$$

Problem 5

Recall that the i th fitted value $\hat{Y}_i$ can be expressed as a linear combination of the response values, i.e.,

$$\hat{Y}_i = \sum_{j=1} h_{ij} Y_j,$$

where

$$h_{ij} = \frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}},$$

and

$$S_{xx} = \sum_{i=1} (x_i - \bar{x})^2.$$

Prove the following properties of the hat-values h_{ij} .

5.i

$$\sum_{j=1} h_{ij}^2 = h_{ii}$$

5.ii

$$\sum_{j=1} h_{ij} x_j = x_i$$

Problem 6

Consider the *regression through the origin model* given by

$$(1) \quad Y_i = \beta x_i + \epsilon_i \quad i = 1, 2, \dots, n \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2).$$

The estimated model at observed point (x, y) is

$$\hat{y} = \hat{\beta} x,$$

where

$$(2) \quad \hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}.$$

Complete the following tasks

6.i Show that

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}$$

is an unbiased estimator of β .

6.ii Compute the standard error of estimator $\hat{\beta}$.

6.iii Identify the probability distribution of estimator $\hat{\beta}$.